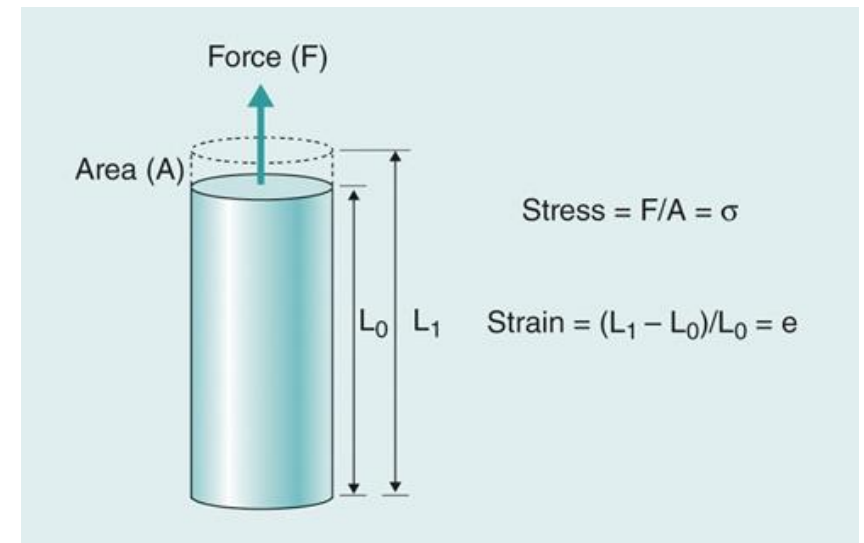


Fundamentals 2 - Mechanical and physical properties

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Stress and Strain

- We can compare materials best through standardized attributes, stress and strain, specific to the material regardless of shape
- Stress - force per unit cross-sectional area
- Strain - fractional change in dimensions caused by force
- Use these values to create stress-strain curves



Types of stress forces

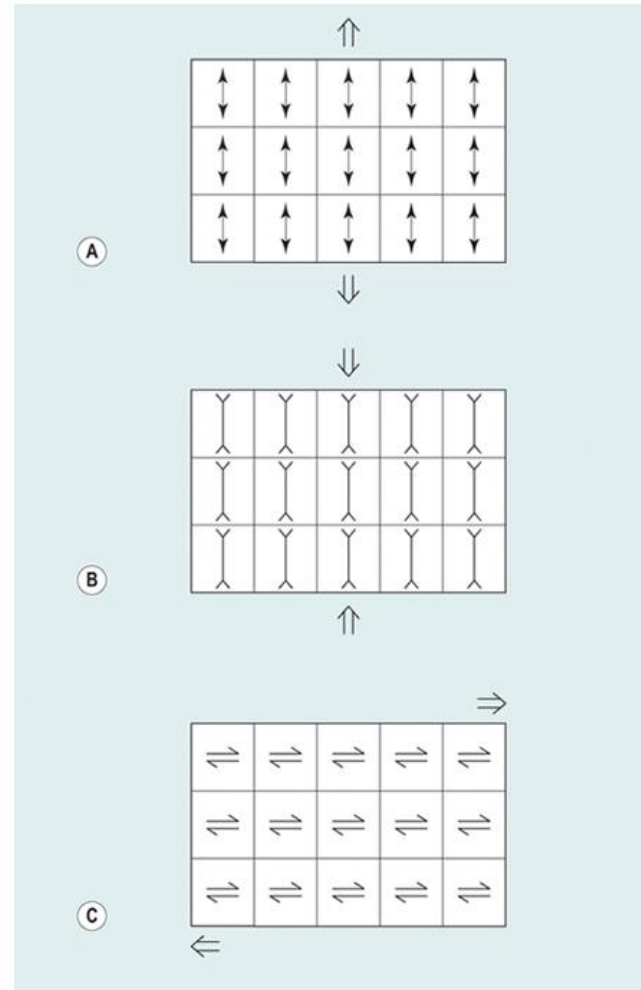


FIGURE 1.6.2 Three principal types of stress: (a) tensile; (b) compressive; (c) shear.

Stress-strain curve

- Linear elastic region – where elastic deformation occurs, material can return to original shape if stress is removed
- Region of plastic deformation - removal of stress does not lead to original shape (marked by yield stress)
- Ultimate tensile strength - max stress a material can withstand
 - Excessive stress and strain above maximum withstanding value --> fracture (fracture strength)
- Ductility - how much strain/elongation a material can withstand prior to fracture (% elongation)

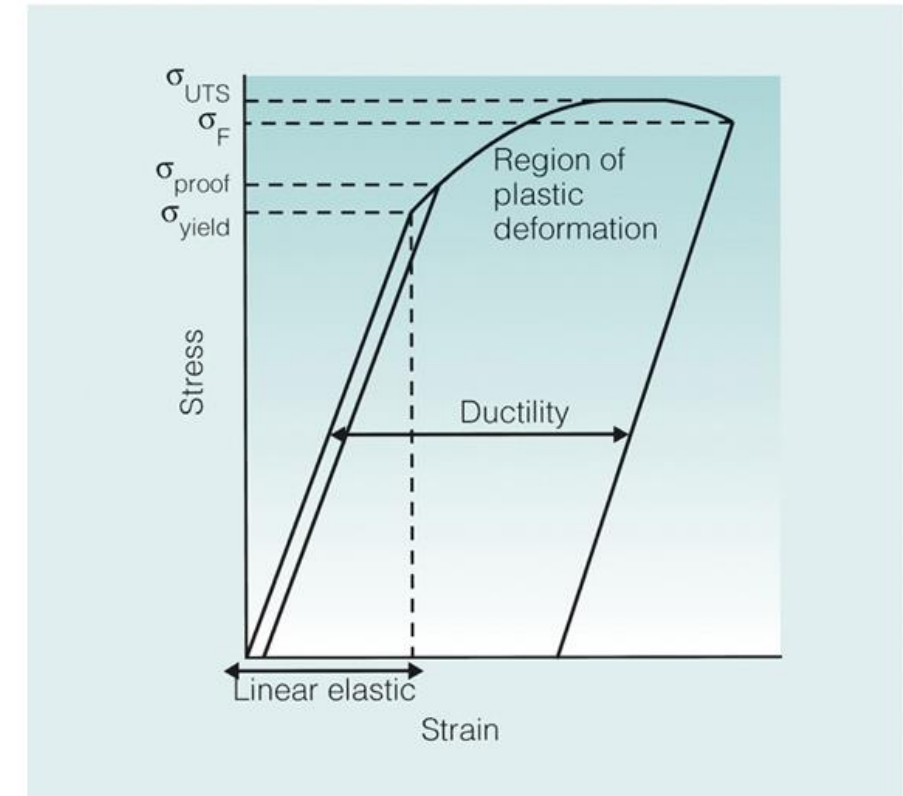


FIGURE 1.6.3 Stress-strain curve for a ductile metal. UTS, Ultimate tensile strength.

Young's modulus of elasticity (Pa)

- Ratio of stress to strain in the linear elastic region – a measure of stiffness of a material

$$\sigma / e = E$$

- In other words, it is the ratio of amount of force needed to create a change in dimension
- What does higher E indicate? What does lower E indicate?
 - Higher E means more force needed to produce dimension change, less elastic
 - Lower E indicates less force needed to produce dimension change, more elastic

Resilience and toughness

- Resilience - amount of energy a material can absorb prior to permanent deformation
 - Area under linear elastic region
- Toughness – amount of energy material can withstand prior to fracture
 - Area under entire stress-strain curve
- Fracture toughness - ability to resist propagation of a fracture crack

TABLE 1.6.1

Fracture Toughness Data for a Variety of Materials

Material	K_{Ic} (MPa m ^{1/2})
Ductile metals	100–350
High-strength steels	50–154
Aluminum alloys	23–45
Wood	11–13
Nylon	3
Porcelain	1

Other mechanical properties

- Hardness – resistance of material to scratching or abrasion
- Fatigue -repeated stress and strain below maximum withstanding value, eventually leading to crack propagation and fracture
- Creep - constant stress (under maximum withstanding value) over time, causing slow/continuous deformation and eventually fracture

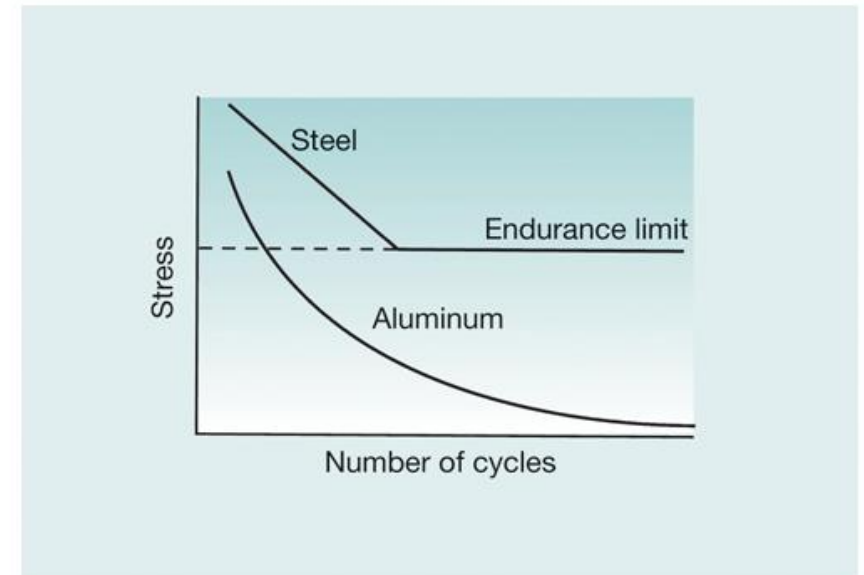


FIGURE 1.6.11 S–N curves for steel and aluminum.

Physical Properties

- Why are they important?
 - Customization of materials – patients are different, teeth are different. To make custom products for the patients, the flow of material should be understood (rheology)
 - Temperature changes in the mouth when eating (thermal properties)
 - Esthetics (optical properties)

Rheology affects handling properties

- Liquids: viscosity
 - Resistance to flow, determined by intermolecular bond strength between neighboring molecules
 - Ex: water (low viscosity) vs. High molecular-weight polymer (entanglement, high viscosity)
 - Some materials (i.e. flowable composite) become less viscous under more shear forces
- Solids: viscoelasticity
 - Viscoelasticity: attributes between a viscous liquid and an elastic solid
 - Duration of force applied on a viscoelastic material determines ability to return to original shape
 - Longer duration of force load --> molecules have more time to rearrange --> higher chance of deformation (why impressions must be removed with a snap-pull)



Thermal properties

- Thermal conductivity - ease of heat transfer through a material (rate of heat flow per unit temperature gradient)
- Thermal diffusivity - how quickly a material internally increases in temperature when transferred heat source
- Specific heat - heat energy required to increase the temperature of a unit volume of material by 1 degree C
- Thermal expansion - energy applied to a material causes miniscule expansion



Coefficient of thermal expansion

- Change in dimension per unit temperature
- For dental restorations, you want the coefficient of thermal expansion to approximate that of teeth
 - Mismatch --> marginal gap formation, adhesive bond degradation
 - Thermal diffusivity determines threshold of heat needed to lead to expansion

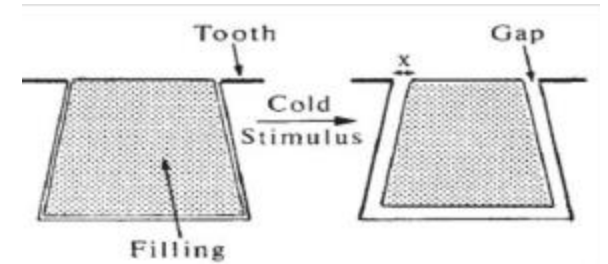


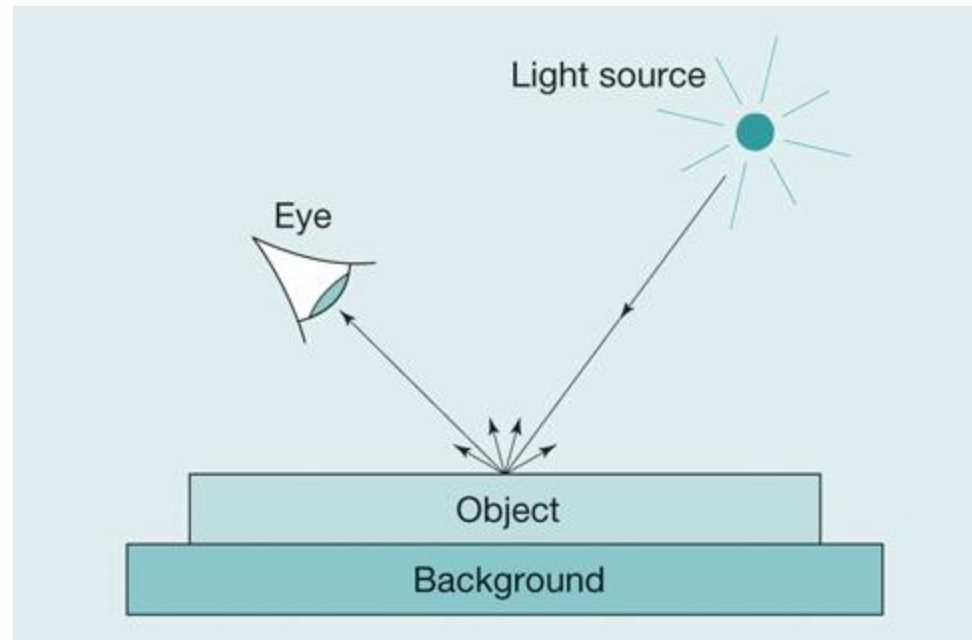
TABLE 1.7.2

Coefficients of Thermal Expansion

Material	α (ppm $^{\circ}\text{C}^{-1}$)
Enamel	12
Dentine	14
Resin composite	20–55
Fissure sealant	80
Porcelain	12
Glass-ionomer cement	8

Optical properties

- Perception is based on light source, reflection off object, and perception by observer



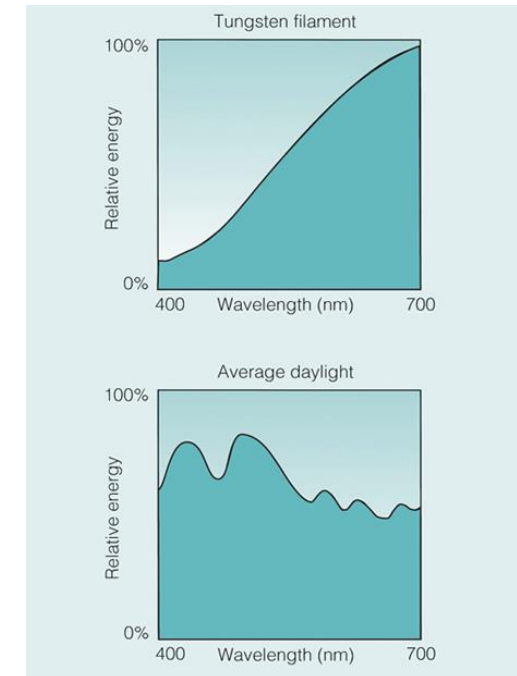
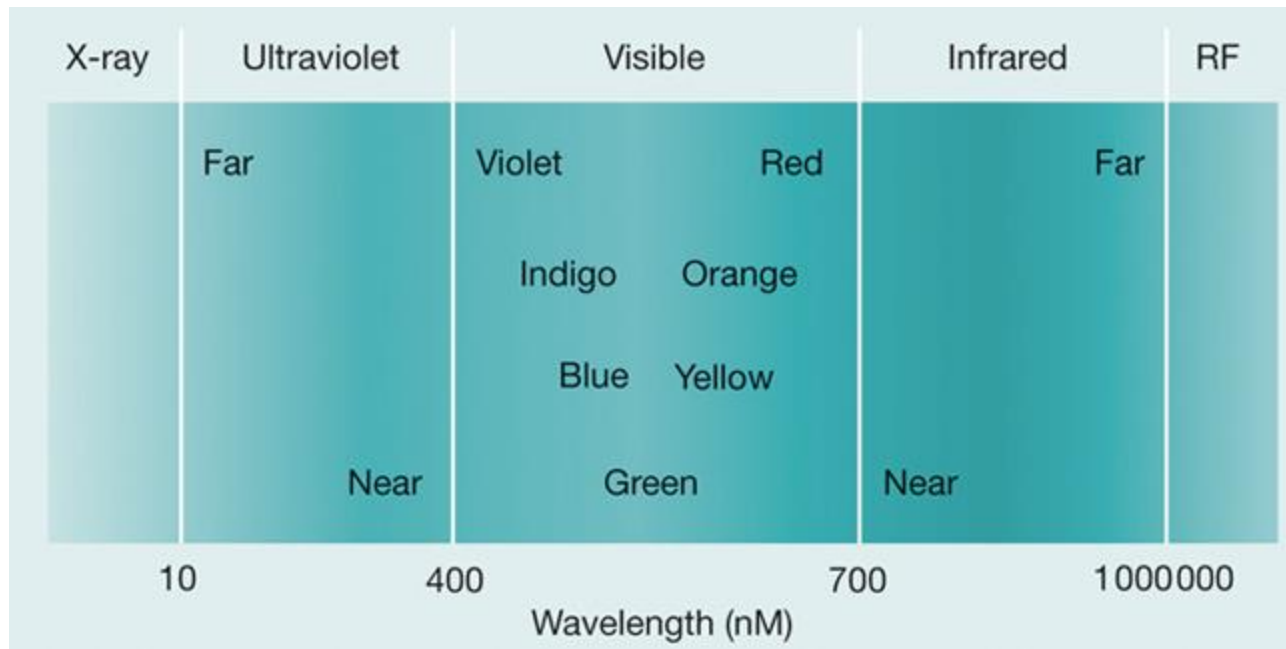
Visual observed characteristics

- Color - function of light source providing a spectrum of light and how the surface of an object transforms this spectrum (based on reflection)
- Translucency - dependent on ability of light to travel through the material, where it may be absorbed, scattered, or reflected
- Surface texture - polished surfaces reflect light well, whereas rough surfaces scatter light (matte appearance)



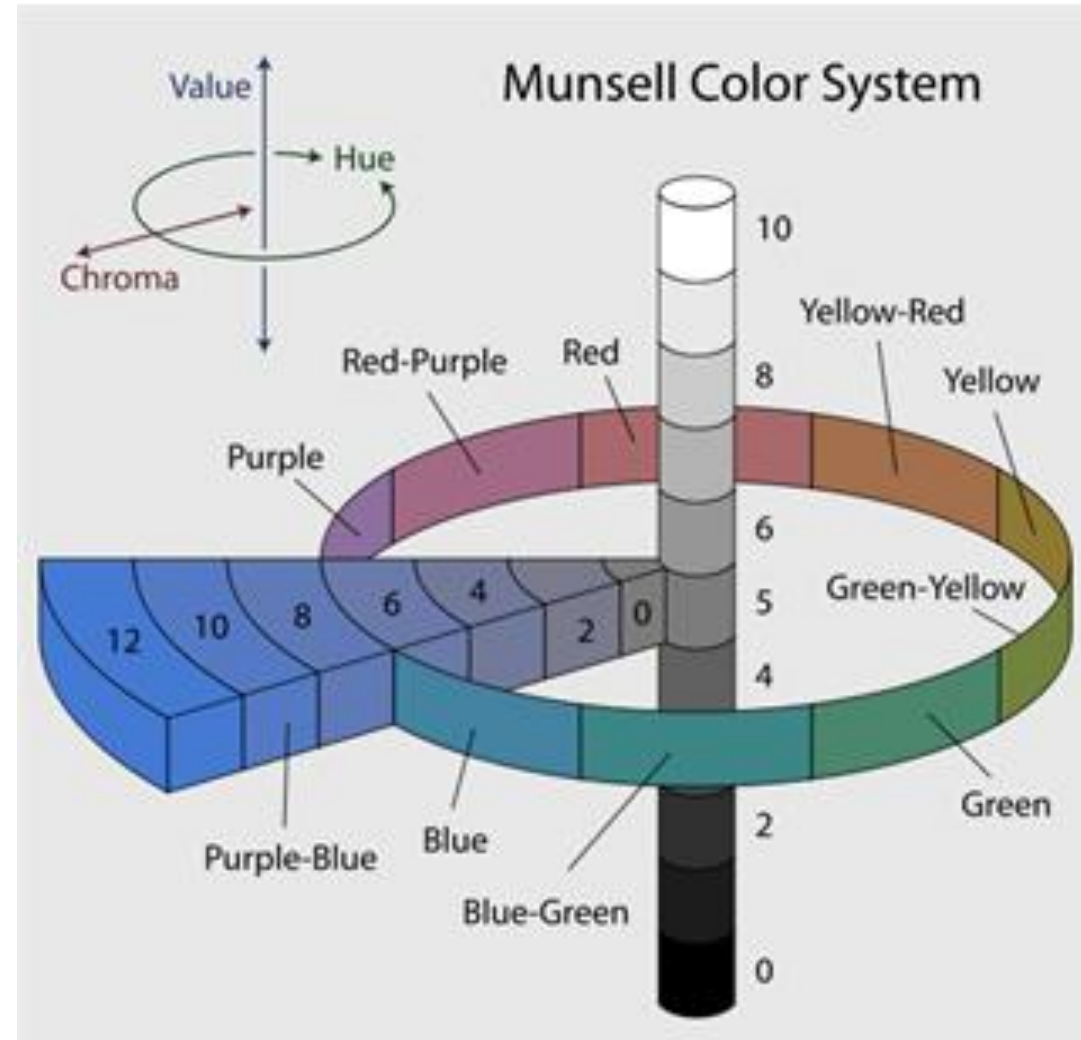
Color

- Generally, color perception is dependent on the individual and highly variable with the source of light reflection



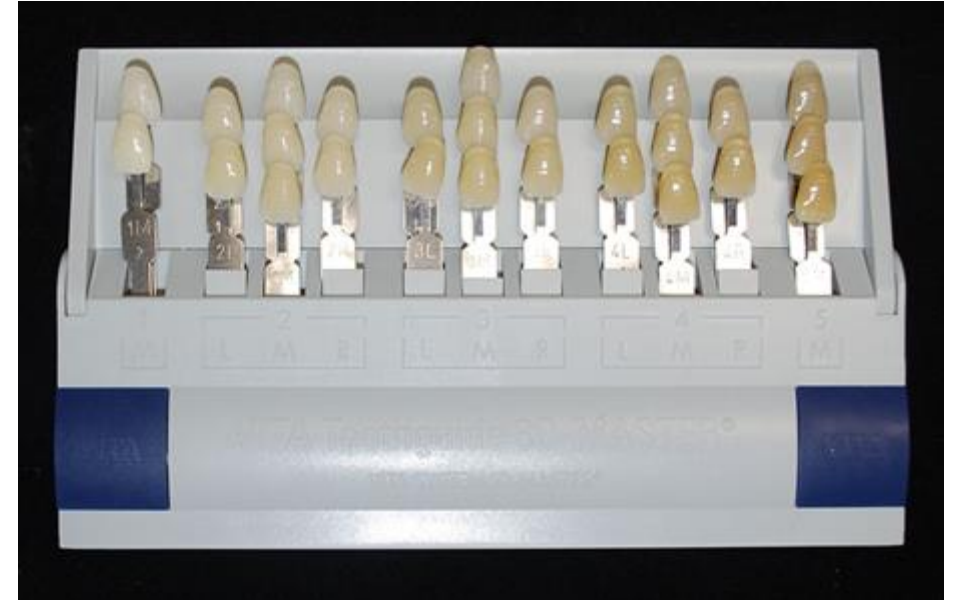
Objectifying color

- Hue: the dominant color (wavelength) of the spectrum of light from the source (ROYGBIV)
- Chroma: strength of the hue; how vivid the color is
- Value: brightness or darkness of an object (black to white/clear)



Color selection

- Value of a color depends on incident light, surface finish of object, and background
- In dentistry, it is best to select colors under natural daylight or a daylight-lamp
 - Metamerism: phenomenon in which objects differ in appearance with different lighting conditions
 - Fluorescence: property of an object allowing absorbing UV light and emitting visible light
 - Enamel is naturally fluorescent, thus restorations may be more noticeable under a fluorescent light source compared to teeth



Translucency

- Transparent (i.e. glass) - light passes through with little distortion
- Translucent - some light passes, some is absorbed, and some is scattered / reflected -> distorted image
- Opaque - absorbs light and reflects / scatters light from surface; color of object depends on which wavelengths of light are reflected and which are absorbed



Surface texture

- Smoother surfaces are shinier because of light reflection
 - Enamel is very smooth, thus reflecting a lot of light directionally
- Rougher surfaces appear more matte because of light scattering
 - Restorations must be smoothed to reflect as much light as enamel

